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Word embeddings: CBOW vs Skip-gram on 1,000 GLP-1 abstracts

Central question: can about 1,000 abstracts learn meaningful word vectors?

Yes, for the main topic of the texts. We used 1,000 abstracts about GLP-1 (177,052 words after removing stopwords). Both models learned that drug names, organs and side effects belong together. They did not learn words that are very rare, or topics the abstracts seldom talk about.

1. How the two models learn

The main idea: words that appear in similar contexts get similar vectors. Each model is a small neural network with two sets of word vectors. We slide a window over each sentence; the window holds c words on each side of a center word.

Skip-gram uses the center word to guess each word around it, so one position gives up to 2c training pairs. CBOW does the opposite: it takes the average of the words around the center and uses it to guess the center word, so one position gives one example. Example (window 2): in "Fasting serum GLP-1 levels were measured by enzyme-linked immunosorbent assay", the center word glp-1 gives Skip-gram 4 pairs (glp-1 → fasting, serum, levels, measured), but gives CBOW only one example (fasting, serum, levels, measured → glp-1).

Checking every word in the vocabulary at each step would be too slow. So both models use negative sampling: they learn to give a high score to the real pair and a low score to a few random "noise" words. We wrote both models ourselves in NumPy (100 dimensions, window 5, 15 epochs) and recorded the loss after every epoch.

2. Why Skip-gram is better for rare and technical words

In Skip-gram, a rare word is the input of every pair it makes. So each time it appears, its own vector is updated several times, and each update moves it toward the words that really appear next to it. In CBOW, the rare word is only one of several words that are averaged together, so its update is small and mixed with the others. When a word appears only a few times, this decides whether its vector is learned or is still mostly random. Our test: when we only changed the random seed, rare words kept 71% of their nearest words in Skip-gram but only 30% in CBOW.

RQ1 · How are CBOW and Skip-gram different?

Training: Skip-gram makes many more updates, so it is slower but learns more from each word. CBOW is faster. Loss: the loss goes down for both, but the two models solve different tasks, so their losses cannot be compared. A low loss also does not mean good word vectors. Nearest words: the models give quite different answers. Skip-gram finds words that appear together with the query (glp-1 → ras, ra, from "GLP-1 RAs" = receptor agonists). CBOW finds words that could replace the query in a sentence (glp-1 → glp1, glp-1/gip). Rare words: Skip-gram handles them well; for CBOW they are mostly noise.

RQ2 · Do the word vectors carry meaning?

We measure how close two words are with cosine similarity, which compares the direction of the two vectors. A cosine number alone can mislead (CBOW gives high scores even to random pairs), so we compared each pair with many random pairs. Related pairs such as semaglutide – liraglutide, nausea – vomiting and kidney – renal clearly beat random pairs in both models, and words from the same group (for example GLP-1 drugs, or stomach side effects) are closer to each other than to other groups. But cancer – tumor and treatment – diagnosis are weak, because these words are side topics in GLP-1 abstracts. So yes, the vectors carry meaning, but only the meaning found in this collection of texts.

RQ3 · How are PCA and t-SNE pictures different?

Both squeeze 100 dimensions into 2 so we can draw the words. PCA is a straight (linear) projection that keeps the directions with the most spread. It is simple and always gives the same result, but two directions keep too little information, so the words form one big cloud (how often a word's nearest words are in its own group: 0.17 in PCA vs 0.67 in the full vectors). t-SNE tries to keep each word next to its nearest neighbors, so it shows clear clusters (0.66). But in t-SNE the distance between clusters and the size of a cluster mean nothing, and the picture changes with its settings. A cluster in a picture is not proof of meaning, so we always checked our claims on the full vectors.

RQ4 · How do the model settings change the results?

Window size has the biggest effect. A small window (2) only sees the next word or two, so it groups words that play the same grammar role. A larger window (5) sees more of the sentence, so it groups words by topic. Changing it replaced most nearest words and made the t-SNE groups weaker, especially for CBOW. Vector size: with this little data, 50 dimensions are already enough; 100 vs 50 changed the results no more than a new random seed. Minimum word count: a higher limit removes noisy rare words, but also removes rare terms we may want.

RQ5 · What are the limits of a small collection?

Every vector is learned only from the sentences we have. With little data, many words are seen just a few times (about half of our vocabulary appears only 2–4 times), so their vectors are uncertain and change with the random seed. Meaning also depends on the topic: here "tzp" means tirzepatide, and cancer is a side topic. Technical terms are split up: different spellings (diarrhea / diarrhoea), different writing (glp-1 / glp1) and multi-word terms (heart failure). Ways to improve: more text, joining multi-word terms, fastText (which uses word parts), and vectors pre-trained on large biomedical text.

Conclusion

A small collection of abstracts is enough to learn useful word vectors for its main topic. Skip-gram is the better choice here because it learns rare technical words well; CBOW is faster and works well for common words.